

DBMS WEEKLY TASK 2

Data Requirements for an E-Commerce Shopping System

Primary Key

A Primary Key is a special attribute in a database table that uniquely identifies every record. It ensures that each row contains a unique value and prevents duplicate entries. A primary key cannot be left empty (NULL) and is essential for connecting related tables in a database.

Example:

If a product has Product_ID = P501, no other product can use the same Product_ID. This allows the system to identify products accurately.

Data Entities Required for the Online Shopping System

1. User

The User entity stores the details of customers using the shopping application.

Attributes:

User_ID (Primary Key)

Full_Name

Email

Mobile_Number

Password

Address

Date_of_Registration

2. Product

Stores information about products available for purchase.

Attributes:

Product_ID (Primary Key)

Product_Name

Brand

Price

Stock_Availability

Product_Image

Description

Category_ID

3. Product Category

Groups products into different sections.

Attributes:

Category_ID (Primary Key)

Category_Name

Display_Order

4. Vendor

Stores information about companies or sellers supplying products.

Attributes:

Vendor_ID (Primary Key)

Vendor_Name

Contact_Number

Email

Location

5. Order Details

Maintains customer purchase records.

Attributes:

Order_ID (Primary Key)

User_ID

Order_Date

Delivery_Address

Order_Total

Order_Status

6. Transaction

Stores payment details.

Attributes:

Transaction_ID (Primary Key)

Order_ID

Payment_Type

Transaction_Date

Payment_Status

Amount_Paid

7. Delivery

Stores shipment and delivery information.

Attributes:

Delivery_ID (Primary Key)

Order_ID

Courier_Service

Tracking_ID

Dispatch_Date

Delivery_Status

8. Customer Feedback

Stores customer ratings and comments.

Attributes:

Feedback_ID (Primary Key)

Product_ID

User_ID

Rating

Comment

Feedback_Date

Supporting Features

Search Option

Customers can search for products using keywords.

Data Required

Search_ID

Keyword

Search_Time

Product Sorting

Products can be arranged in different ways.

Sort Methods

Popularity

New Arrivals

Price (Ascending)

Price (Descending)

Highest Rated

Product Filters

Customers can refine search results using filters.

Available Filters

Brand

Category

Price Range

Discount

Color

Size

Availability

Gender Classification

Fashion and accessories can be classified based on gender.

Options

Men

Women

Boys

Girls

Unisex

Offers and Discounts

Stores promotional details.

Attributes

Offer_ID

Offer_Name

Discount_Percentage

Start_Date

End_Date

Shopping Wishlist

Allows users to save products for future purchase.

Attributes

Wishlist_ID

User_ID

Product_ID

Added_Date

Conclusion

An Online Shopping System requires several entities to store customer, product, vendor, order, payment, and delivery information. Additional features such as product search, sorting, filtering, gender classification, offers, and wishlists improve the user experience. The use of Primary Keys ensures that each record is unique and helps establish relationships between different database tables.